MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 14: Series representation of complex functions

Nick Strigul
Washington State University Vancouver
nick.strigul@wsu.edu

February 27, 2023

Lecture outline

1. Series representation of functions
2. Taylor formula theorem
3. Examples
4. Isolated Zeros Theorem
5. Converging Sequence of Zeros Theorem
6. Identity Theorem

Series representation of differentiable functions

Remark: We have seen that a power series that converges in an open disk defines an infinitely differentiable function in that disk. We now go the other way and show that a function that is differentiable in an open disk can be represented as a power series expanded about the center of that disk.

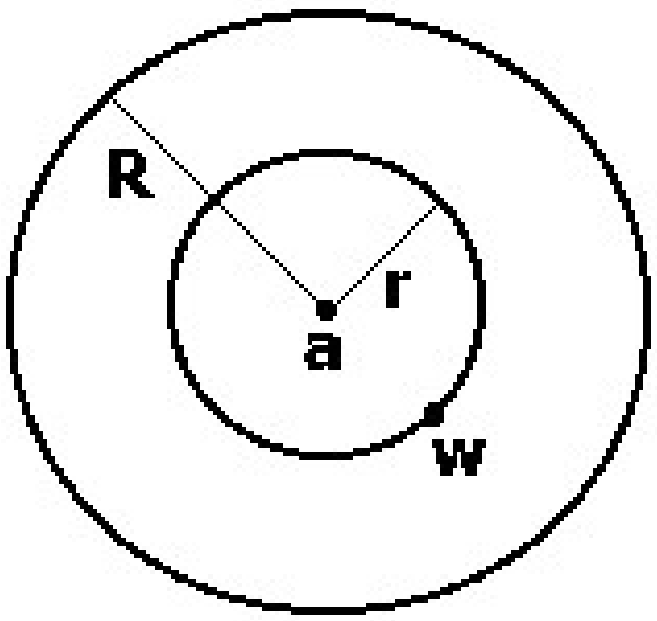
Theorem

Taylor Series. Let f be differentiable in an open disk D about a. Then for each $z \in D$

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (z-a)^n$$

Proof of the Taylor Series Theorem. Part 1

Let z be any point in D, and let R be the radius of D. Let r be such that $0 < r < R$ and let $|z - a| < r$, and let γ be the circle of radius r about a.



Proof of the Taylor Series Theorem. Part 2

By the Cauchy integral formula, with w as the variable of integration on γ we have

$$f(z) = \frac{1}{2\pi i} \oint_\gamma \frac{f(w)}{w-z} dw$$

We can write

$$\frac{1}{w-z} = \frac{1}{w-a+a-z} = \frac{1}{w-a-(z-a)} = \frac{1}{w-a} \frac{1}{1-\dfrac{z-a}{w-a}}$$

Now w is on γ and z is in the interior of γ so that

$$\left| \frac{z-a}{w-a} \right| < 1.$$

Thus (using geometric series formula in reverse)

$$\frac{1}{w-z} = \frac{1}{w-a} \sum_{n=0}^{\infty} \left(\frac{z-a}{w-a} \right)^n = \sum_{n=0}^{\infty} \frac{(z-a)^n}{(w-a)^{n+1}}$$

Proof of the Taylor Series Theorem. Part 3

Hence

$$f(z) = \frac{1}{2\pi i} \oint_\gamma \frac{f(w)}{w-z} dw = \frac{1}{2\pi i} \oint_\gamma f(w) \left[\sum_{n=0}^{\infty} \frac{(z-a)^n}{(w-a)^{n+1}} \right] dw$$

Termwise integration is possible since the series is uniformly convergent with respect to w because

$$\left| \frac{(z-a)^n}{(w-a)^{n+1}} \right| = \frac{|z-a|^n}{r^{n+1}} < 1$$

Thus

$$f(z) = \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \oint_\gamma \frac{f(w)}{(w-a)^{n+1}} dw \right) (z-a)^n = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (z-a)^n$$

Proof of the Taylor Series Theorem. Part 4

Remark (uniqness): If $\sum c_n (z-a)^n = \sum d_n (z-a)^n$, in an open disk then about a then $c_n = d_n \ \forall \ n$. The reason is that if f is the function defined by these power series, then

$$c_n = d_n = \frac{f^{(n)}(a)}{n!}$$

This means that whatever legitimate means is used to find a power series representation about a this must be the Taylor series of that function about a.

Example 1

Example

Find the MacLaurin series (Taylor series about the origin) for e^z, $\sin z$, and $\cos z$.

Since

$$\frac{d^n e^z}{dz^n} = e^z$$

for all n and $e^0 = 1$, then

$$e^z = \sum_0^\infty \frac{z^n}{n!}.$$

From the definition of $\cos t$ we have

$$\cos t = \frac{e^{it} + e^{-it}}{2} = \frac{1}{2}\sum_{n=0}^\infty \frac{(it)^n + (-it)^n}{n!} = \sum_{m=0}^\infty \frac{(-1)^m t^{2m}}{(2m)!}$$

whereas

$$\sin t = -\frac{d}{dt}\sum_{m=0}^\infty \frac{(-1)^m t^{2m}}{(2m)!} = \sum_{m=0}^\infty \frac{(-1)^{m+1} t^{2m+1}}{(2m+1)!}$$

Example 2

Example

Expand $\dfrac{2i}{4+iz}$ in a Taylor series about $-3i$.

We want a Taylor series of the form $\sum c_n (z+3i)^n$.

$$\frac{2i}{4+iz} = \frac{2i}{4+i(z+3i)+3} = \frac{2i}{7+i(z+3i)} = \frac{2i}{7}\frac{1}{1+(i/7)(z+3i)}$$

However, the geometric series implies $\dfrac{1}{1+t} = \dfrac{1}{1-(-t)} = \sum_0^\infty (-1)^n t^n$ if $|t| < 1$.

Letting $t = (i/7)(z+3i)$ implies

$$\frac{2i}{4+iz} = \frac{2i}{7}\sum_{n=0}^\infty (-1)^n \left(\frac{i}{7}(z+3i)\right)^n = 2\sum_{n=0}^\infty (-1)^n \frac{i^{n+1}}{7^{n+1}}(z+3i)^n.$$

This series converges for $|(i/7)(z+3i)| < 1$, that is for $|z+3i| < 7$.

More on Taylor formula

Remark: The existence of a power series expansion of a function implies the existence of an antiderivative.

Corollary

Let f be differentiable in an open disk about a. Then there is a differentiable function g such that $g'(z) = f(z)\ \forall\ z$ in this disk.

Proof. We know that

$$f(z) = \sum_{n=0}^{\infty} c_n (z-a)^n$$

in this disk about a. Define

$$g(z) = \sum_{n=0}^{\infty} \frac{1}{n+1} c_n (z-a)^{n+1}$$

for z in the disk. It is easy to check that the series for $g(z)$ has the same radius of convergence as the series for $f(z)$. The power series for $g(z)$ can be differentiated term by term, which yields $g'(z) = f(z)$.

Isolated Zeros Theorem

Remark: Taylor series yield information about the zeros of differentiable functions. We say a number ζ is a *zero* of f if $f(\zeta) = 0$. A zero ζ is *isolated* if there is an open disk about ζ having no other zero of f than ζ. For example, the zeros of $\sin z$ are isolated and occur at $n\pi$ for any integer n.

We shall see that the zeros of a differentiable function must be isolated if the function is not simply the zero function. First we establish

Theorem

Isolated Zeros: Let f be differentiable on an open set G and let ζ be a zero of f in G. Then either ζ is an isolated zero, or there is a disk about ζ on which $f(z)$ is identically zero.

Proof of Isolated Zeros Theorem

Proof: We may write

$$f(z) = \sum_{0}^{\infty} c_n (z-\zeta)^n$$

in some open disk D about ζ. We consider two cases:

1. $c_n = 0 \ \forall \ n$, so that $f(z) \equiv 0$ throughout D.
2. Some coefficient in the series is not zero. Let m be the smallest integer such that $c_m \neq 0$. For $z \in D$

$$f(z) = \sum_{n=m}^{\infty} c_n (z-\zeta)^n = \sum_{n=0}^{\infty} c_{n+m} (z-\zeta)^{n+m} = (z-\zeta)^m \sum_{n=0}^{\infty} c_{n+m} (z-\zeta)^n$$

Now

$$\sum_{n=0}^{\infty} c_{n+m} (z-\zeta)^n$$

is a power series that defines a differentiable function $g(z)$ in D and thus $f(z) = (z-\zeta)^n g(z)$. Now $g(\zeta) = c_m \neq 0$, so there is some open disk K about ζ in which $g(z) \neq 0$. Thus, in K, $f(z) \neq 0$ if $z \neq \zeta$, so that ζ is an isolated zero of f.

Converging sequence of zeros

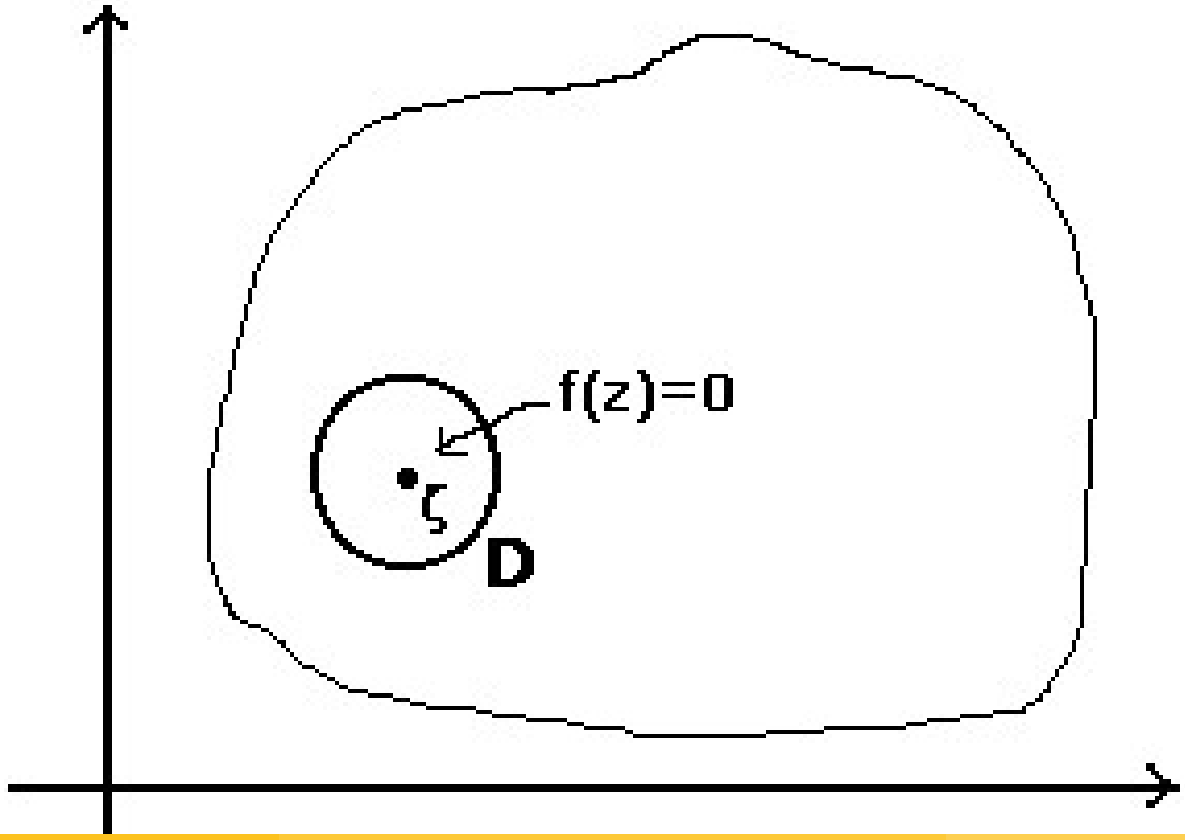
Remark: We shall now use this theorem to show that if f is a differentiable function that vanishes on a convergent sequence of points in a domain, then f must be identically zero throughout the entire domain. This is surprising because it means that values of the function throughout the domain can be determined by their values on a convergent sequence of points which may lie in a small part of the domain.

Theorem

Let f be differentiable on a domain G. Suppose that $\{z_n\}$ is a sequence of zeros of f in G converging to a point in G. Then $f = 0 \ \forall \ z \in G$.

Proof - Converging sequence of zeros theorem. Part 1

Proof: Recall that a domain is an open, connected set.
Suppose that $z_n \to \zeta$ where $\zeta \in G$. Then $f(z_n) \to f(\zeta)$. But since $f(z_n) = 0$ for each z_n, then $f(\zeta) = 0$. Also, since a sequence of zeros converges to ζ, we see that ζ is not an isolated zero of f. By the previous theorem there is an open disk D about ζ in which $f(z) \equiv 0$.

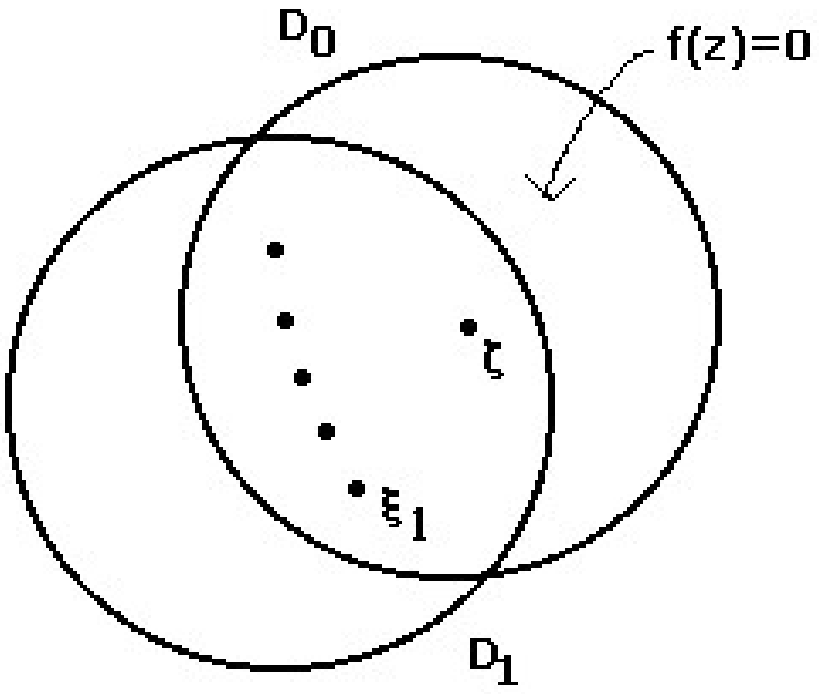


Proof - Converging sequence of zeros theorem. Part 2

We want to show that this forces $f(z)$ to be zero throughout G. Let w be any point of G. We shall show $f(w) = 0$.
Now G is connected so there is a path Γ in G from ζ to w. Choose a number r such that every point of Γ is at a distance of at least $2r$ from any boundary point of G, and also choose r to be less than the radius of D. Now choose a succession of points $\zeta = \xi_0, \xi_1, \ldots, \xi_n = w$ on Γ so that the distance between successive ξ_j, s does not exceed r. Form an open disk D_j of radius about ξ_j. By the choice of r each ξ_j is in D_{j-1}, D_j, and D_{j+1} for $j = 1, \ldots, n-1$. Also, $\xi_0 = \zeta$ is in D_0 and in D_1 and $\xi_n = w$ is in D_{n-1} and D_n.

Proof - Converging sequence of zeros theorem. Part 3

Since ξ_1 is also in D_0, on which $f(z) = 0$, we can find a sequence of points common to D_0 and D_1 converging to ξ_1. But $f(z) = 0$ throughout D_0, so $f(z) = 0$ at each point of this sequence; hence $f(z) = 0$ throughout D_1 by the preceding theorem.



Proof - Converging sequence of zeros theorem. Part 4

ξ_2 is in D_1 and D_2. Form a sequence of points common to D_1 and D_2 converging to ξ_2. Since $f(z)$ vanishes on D_1 then $f(z) = 0$ at each point of this sequence, so $f(z) = 0$ throughout D_2.
Continuing in this way we "walk" along Γ through the $\xi_j's$ getting $f(z) = 0$ on each open D_j in turn, until we finally get $f(z) = 0$ on D_n. But w is in D_n so $f(w) = 0$, and this completes the proof.

Identity Theorem

Theorem

Identity Theorem. Let f and g be differentiable on a domain G. Suppose $\{z_n\}$ is a sequence in G that converges to a point of G, and that $f(z_n) = g(z_n)$ for each n. Then $f(z) = g(z)\ \forall\ z \in G$.

Proof: Apply the preceding theorem to $f(z) - g(z)$.

Thank you